

Untrained CNNs ~~Match Backpropagation at~~ Exceed Backpropagation in V1 Alignment at High Evaluation Resolution: A Systematic RSA Comparison of Four Learning Rules Against Human fMRI

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Abstract

~~Correction Revised version (August 2026v4). An evaluation-mode defect affected the predictive-coding and STDP conditions of this study; those results should not be used pending re-computation. The random, backpropagation and feedback-alignment conditions, which carry the untrained-versus-trained claim, are unaffected. See the correction note following this abstract.~~
This version replaces the noise ceiling, restricts all analyses to stimulus pairs that were presented in different fMRI runs, and uses the repaired evaluation of all five conditions; see the revision note below.

A central question in computational neuroscience is whether the learning rule used to train a neural network determines how well its internal representations align with those of the human visual cortex. We present a systematic comparison of four learning rules—backpropagation (BP), feedback alignment (FA), predictive coding (PC), and spike-timing-dependent plasticity (STDP)—applied to identical convolutional architectures and evaluated against human fMRI data from the THINGS-fMRI dataset (720 stimuli, 3 subjects) using Representational Similarity Analysis (RSA). All models process stimuli at 224×224 resolution; results are averaged across 5 random seeds, and uncertainty is estimated by resampling stimuli. Crucially, we include an untrained random-weights baseline that ~~reveals the dominant role isolates the contribution~~ of architecture. At V1/V2, the untrained baseline *exceeds* backpropagation ($p=0.076$ – $p=0.053$ vs. $p=0.034$; $\Delta p=+0.044$, $p<0.001$; $p=0.022$; $\Delta p=+0.031$, 95% CI [0.019, 0.043]), and STDP achieves the highest V1 alignment among trained rules ($p=0.064$) the same holds at V2. This advantage depends on the evaluation resolution: the models are trained at 32 px and evaluated at 224 px. In the 5-seed model set of the resolution study (same architecture and training; gap +0.030 at 224 px), the gap vanishes when the models are evaluated at 32 px ($\Delta p=-0.001$, 95% CI [-0.011, 0.009]; arXiv:2608.12408). At LOC, only BP reliably exceeds BP, PC and STDP exceed the

random baseline ($p=0.012$ vs. $p=0.005$, $p<0.001$). FA is borderline, and the trained rules do not differ from each other. At IT, all five conditions converge ($p=0.008$ – 0.014) with no significant pairwise differences among trained rules ($p>0.05$, FDR-corrected). FA consistently no two conditions can be distinguished; the reliability of the IT data is low (leave-one-subject-out lower bound 0.021). FA produces the lowest alignment at V1, V2, and LOC ($p=0.012$). Only the untrained network reaches the lower noise bound at V1, below all other conditions. Partial RSA confirms all effects survive pixel-similarity control. Seed variability is small relative to between-rule differences at V1; at V2 it and STDP are not distinguishable from the bound, and every condition lies below it at LOC and IT. These results ~~demonstrate that~~ indicate that, at the 224 px evaluation resolution used here, early visual alignment is architecture-driven, learning rules differentiate driven more by the architecture than by the learning rule, and that learning-rule differences are resolvable only at intermediate areas, and all rules converge at the highest levels of the hierarchy.

Keywords: representational similarity analysis, visual cortex, biologically plausible learning, random baseline, STDP, feedback alignment, predictive coding, fMRI, THINGS

~~Correction~~ Revision note (August 2026v4)

~~A defect affecting two of five conditions.~~ In the implementation used for this study, the predictive-coding and STDP model classes override `eval()` with a no-op. Feature extraction calls `eval()` before computing activations, so for those two conditions the batch-normalisation layers remained in training mode and normalised each evaluation batch by its own statistics, while the random, backpropagation and feedback-alignment conditions used their stored running statistics. This is a rule-correlated difference in the evaluation pipeline, not in Earlier versions contained the

following errors. (i) The noise ceiling was described as split-half reliability with a Spearman-Brown correction; with three subjects that is not a valid estimate, and the learning rules.

Repairing the defect and re-running the full-printed bounds could not be reproduced from any stored result. It is replaced by the leave-one-subject-out lower bound of Nili et al. (2014); the upper bound is not informative at $N = 3$ and is reported only in the appendix. (ii) Subjects sub-01 and sub-02 saw the stimuli in identical order, so pairs of stimuli from the same run carry a run component that is common to these two subjects; it inflated their agreement and the lower bound at LOC and IT. All analyses now use only the 210,205 of 258,840 stimulus pairs (81.2%) that lie in different runs for every subject. (iii) The PC and STDP conditions were evaluated with batch normalisation left in training mode. The correction note of v3 (August 2026) identified this defect, but the values reported in the paper were not updated; all values are now from the repaired five-seed design leaves the random, backpropagation and feedback-alignment conditions unchanged to within $\Delta\rho \leq 0.0013$ at re-run. (iv) Several values were rounded twice (e.g. 0.075 printed as 0.076), and the V1 across six evaluation resolutions, and reproduces the untouched random condition bit-identically. It changes the two affected conditions substantially. At V1 and 224px, predictive coding falls from $\rho = 0.056$ to $\rho = 0.016$ and STDP from $\rho = 0.064$ to $\rho = 0.037$.

The following claims of this paper are therefore *not supported* and should be disregarded: that STDP achieves the highest V1 alignment among trained rules; that predictive coding and STDP lead among trained rules at V1/V2; and the associated discussion (“PC and STDP as biologically plausible routes to V1 alignment”) and conclusion. The reported values for random weights, backpropagation and feedback alignment, and the FDR-corrected comparisons among them, are unaffected.

On the untrained baseline. The untrained condition’s batch-normalisation layers hold their initialised statistics, so in evaluation mode they act as the identity, whereas every trained condition carries statistics accumulated during training. This is a difference between the conditions that the paper did not report. A 2×2 calibration control varying the calibration set and the calibration resolution, with the convolutional weights held bit-identical, indicates that it does not produce the result: with statistics calibrated at the evaluation resolution the untrained network still exceeds backpropagation at V1 by +0.041 (gap was quoted from a different run (appendix, Table 12)). (v) Two statements about LOC contradicted each other. (vi) The hierarchy figure showed, for each layer, the mean over all four ROIs, contrary to its caption, which described the fixed layer-to-ROI mapping; it now

shows that mapping. (vii) The untrained network was described as He-normal initialised; it uses PyTorch’s default `kaiming_uniform_`, while FA and STDP use `kaiming_normal_`. (viii) The accuracy table reported accuracy on training batches; it now reports CIFAR-10 statistics) or +0.033 (statistics from a disjoint pool of THINGS images), 5/5 seeds. The central result of this paper is not an artefact of an unnormalised baseline. It is, however, strongly dependent on test accuracy. What changes: the lower bounds are 0.051, 0.024, 0.018 and 0.021 (V1 to IT); the claim that the evaluation resolution, which this paper held fixed at 224px; see arXiv:2608.12408.

On the noise-ceiling argument. Section 4.2 argues that because all conditions lie within or near the split-half noise ceiling, they models “exhaust most of the available signal. That is true and less reassuring than it reads: a single scalar luminance value per image reaches $\rho = 0.075$ against the same” is withdrawn; PC no longer exceeds BP at V1RDM, essentially matching, and STDP exceeds it only in the per-subject convention; IT results are stated as not distinguishable at low reliability; statistics are stimulus-bootstrap CIs instead of permutation tests. What holds: the untrained network exceeds BP at V1 and V2, BP exceeds the untrained network’s 0.076 (arXiv:2608.12408). The luminance figure involves no model and no readout, so it bounds the brain side directly; at LOC, and FA has the lowest alignment at V1. FA at V2 is no longer a primary comparison, because its ordering against BP and PC depends on the layer assigned to V2 (appendix, Table 6); reducing the family of primary comparisons from 11 to 7 does not change any significance decision (appendix, Table 7). Still on the original state are the filter and Gabor analysis (weights of the original model set; unaffected by the defect), the models’ figures are specific to rank correlation on globally pooled features. Occupying most of the resolvable signal at V1 under this comparison is a low bar.

Two smaller corrections. Section 3.3 states that the random condition is initialised with He-normal weights. The implementation uses PyTorch’s default `kaiming_uniform_`; feedback alignment and STDP do call `kaiming_normal_`, so the initialisation scale differs by roughly a factor of 2.4 at Conv1 training curves, and the single untrained condition is the matched control for three of the five conditions rather than all five. Separately, Limitation (8) raised the possibility that evaluating 32-px-trained models at 224-px contributes to the random baseline’s V1 advantage; that limitation is now quantified as a main effect in arXiv:2608.12408, and the mechanism it proposed is refuted there batch-normalisation calibration control (all stimulus pairs, mean-RDM convention); test accuracy is available for seed 0 only.

A fully re-computed version of the affected analyses is in preparation. The repaired implementation and all

~~re-run results are at .~~

1 Introduction

Deep neural networks (DNNs) trained with backpropagation have become the dominant computational models of the ventral visual stream (Yamins and DiCarlo, 2016). A well-established finding is that the hierarchy of features learned by DNNs mirrors the hierarchy of representations from V1 through IT cortex, as quantified by representational similarity analysis (RSA) and related metrics (Kriegeskorte et al., 2008; Schrimpf et al., 2020). This correspondence has been interpreted as evidence that the objectives and inductive biases of deep learning capture something fundamental about visual processing.

Yet backpropagation is neurobiologically implausible in at least three well-known respects: it requires symmetric forward and feedback weights (the weight transport problem), it relies on a global error signal propagated backward through the network, and the forward and backward passes must be temporally separated (Crick, 1989; Lillicrap et al., 2020). These observations have motivated work on biologically plausible alternatives, including feedback alignment (Lillicrap et al., 2016), predictive coding (Rao and Ballard, 1999; Whittington and Bogacz, 2017), and spike-timing-dependent plasticity (Bi and Poo, 1998).

A key question is: does the choice of learning rule matter for brain alignment? If alignment arises from the architecture and objective rather than the weight-update rule, then alternative rules—or even no training at all—should yield comparable alignment. Prior work has compared individual learning rules to brain data (Schrimpf et al., 2020), but to our knowledge no study has systematically compared all four rules together with an untrained random-weights baseline on the same architecture, stimuli, and RSA pipeline.

We address this gap with five conditions: BP, FA, PC, STDP, and a random-weights (untrained) baseline, all applied to identical CNNs trained on CIFAR-10, with layer-wise RDMs compared to human fMRI from the THINGS-fMRI dataset.

Our main contributions are: (1) a head-to-head RSA comparison of four learning rules plus a random baseline at 224×224 resolution, averaged across 5 seeds; (2) the finding that ~~V1/V2 alignment is architecture-driven, with, at 224 px evaluation,~~ the untrained CNN ~~exceeding~~ exceeds BP at V1 ($\Delta\rho = +0.044, p < 0.001$) and V2 (V1: $\Delta\rho = +0.031, 95\% \text{ CI } [0.019, 0.043]$), while at V1 the two do not differ at the 32 px training resolution; (3) the result that at IT, ~~all trained rules converge and are statistically indistinguishable from each other~~ no two conditions can be distinguished at the low reliability of the IT data; (4) the finding that ~~only LOC shows a reliable learning effect, with BP being the sole rule to~~

~~significantly exceed the random baseline~~ LOC is the only area where trained rules exceed the untrained baseline: BP, PC and STDP do so while the trained rules do not differ from each other; and (5) the ~~consistent~~ finding that FA produces the lowest brain alignment at V1, V2, and LOC.

2 Related Work

RSA as a model evaluation tool. Kriegeskorte et al. (2008) introduced RSA as a framework for comparing representations across measurement modalities. The core idea is to represent each system’s response to a stimulus set as a pairwise dissimilarity matrix (RDM) and to assess model–brain alignment as the rank correlation between RDMs. RSA has since become a standard tool for comparing DNNs to neural data (Yamins and DiCarlo, 2016; Schrimpf et al., 2020).

DNNs and the visual hierarchy. Task-optimized DNNs reproduce the representational hierarchy of the ventral visual stream, with early layers matching V1 and deeper layers matching IT (Yamins and DiCarlo, 2016). The Brain-Score benchmark (Schrimpf et al., 2020) systematizes this evaluation across architectures and datasets. Most high-scoring models are trained with standard BP on ImageNet. Importantly, Schrimpf et al. (2020) noted that untrained networks retain some brain alignment, but did not systematically compare this against multiple learning rules.

Biologically plausible learning rules. Feedback alignment (Lillicrap et al., 2016) replaces the transpose weight matrix in the backward pass with fixed random matrices, removing the weight transport problem. Lillicrap et al. (2016) showed that FA still allows learning in fully-connected networks, though its effectiveness in convolutional architectures is less established. Predictive coding (Rao and Ballard, 1999) frames inference as hierarchical prediction-error minimization; Whittington and Bogacz (2017) showed that PC implements an approximation to BP under certain conditions, with learning driven entirely by local Hebbian updates. Spike-timing-dependent plasticity (Bi and Poo, 1998) updates synaptic weights based on the relative timing of pre- and postsynaptic spikes, and is one of the best-characterized forms of synaptic plasticity in mammalian cortex. Each of these rules has been studied individually in the context of neural representation learning, but direct comparisons on identical architectures and evaluation pipelines are rare.

Random networks and early visual cortex. Saxe et al. (2011) demonstrated that random-weight networks can exhibit non-trivial visual structure. Concurrently with our work, Truzzi and Cusack (2025) showed that random-weight CNNs predict early visual cortex responses when model complexity is sufficient. Our work extends these observations by systematically comparing four distinct learning rules against

the random baseline, and by demonstrating that the architecture-dominance effect is not limited to V1: at IT, all rules—including random—converge with no significant pairwise differences across the visual hierarchy.

Sparse coding and V1. Olshausen and Field (1996) showed that unsupervised learning of a sparse code for natural images produces receptive fields resembling V1 simple cells, establishing that correlation-based learning from image statistics suffices to produce V1-like representations without supervised objectives.

3 Methods

3.1 Dataset

We used the THINGS-fMRI dataset (Gifford and Cichy, 2022), comprising 7T fMRI responses to images, part of THINGS-data (Hebart et al., 2023): 3T fMRI responses (Siemens Magnetom Prisma) of 3 subjects to 8,740 unique images of 720 object concepts from the THINGS object concept database (Hebart et al., 2019) database (Hebart et al., 2019), recorded in 12 main-experiment sessions of 10 runs each. We selected 720 stimuli (one image per concept, each presented once). fMRI responses were available for 3 subjects across 4 regions of interest (ROIs): V1, V2, LOC, and IT. RDMs were computed from averaged trial responses using correlation distance, then averaged across subjects to produce a single mean-brain RDM per ROI. For each subject and ROI, single-trial responses were z-scored per voxel across all trials, and a subject RDM was computed as the correlation distance between the response patterns of the 720 stimuli.

3.2 Model Architecture

All five conditions used an identical CNN architecture consisting of three convolutional layers (Conv1–Conv3, each followed by BatchNorm, ReLU, and 2×2 max-pooling) and two fully connected layers (FC1, FC2). The output layer uses softmax for 10-class classification on CIFAR-10. Training used 8,000 CIFAR-10 samples for 40 epochs at native 32×32 resolution. For RSA evaluation, THINGS stimuli were resized to 224×224 and passed through the same trained (or untrained) network; this is standard practice in model–brain comparisons where training and evaluation datasets differ (Schrumpf et al., 2020). All results are from the five-seed re-run with the repaired evaluation mode (Revision note, item iii); the original model set is reported as a sensitivity column in the appendix.

The fixed layer-to-ROI mapping follows the standard ventral-stream correspondence: is Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT. V1 and V2 are both mapped to Conv1; Conv2 is not used in the

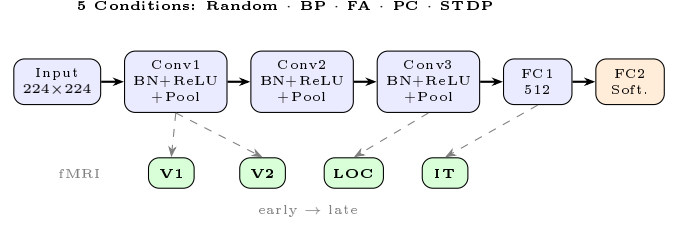


Figure 1: Model architecture and layer-to-ROI mapping. Models are trained on CIFAR-10 (32×32); for RSA, THINGS stimuli are resized to 224×224 at feature extraction time. Dashed arrows: Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT.

fixed mapping (its values are given in the best-layer analysis, Table 3).

3.3 Learning Rules

Random Weights serves as the architectural baseline. The network is initialized with He-normal-weights PyTorch’s default kaiming_uniform, and never trained, isolating the contribution of the convolutional architecture (filters, ReLU, pooling) to brain alignment. FA and STDP initialize with kaiming_normal, whose standard deviation is larger by a factor of about 2.4. The untrained network therefore shares its initialization with BP and PC and is their matched control; for FA and STDP, comparisons with the untrained network also include the difference in initialization scale.

Backpropagation (BP) serves as the supervised baseline: $\Delta W_l = -\eta \frac{\partial \mathcal{L}}{\partial W_l}$.

Feedback Alignment (FA) (Lillicrap et al., 2016) replaces $W_l^\top$ in the backward pass with a fixed random matrix B_l :

$$\Delta W_l = \delta_l^{\text{FA}} \mathbf{h}_{l-1}^\top, \quad \delta_l^{\text{FA}} = f'(z_l) \odot (B_{l+1} \delta_{l+1}^{\text{FA}}). \quad (1)$$

Predictive Coding (PC) (Rao and Ballard, 1999; Whittington and Bogacz, 2017) minimizes a hierarchical prediction-error energy $F = \sum_l \|\varepsilon_l\|^2$ through iterative inference ($T_{\text{inf}} = 10$ steps, inference rate $\alpha = 0.02$):

$$\Delta W_l \propto \varepsilon_l \mathbf{x}_{l-1}^\top. \quad (2)$$

Prediction weights $\mathbf{P}_l$ (implemented as transposed convolutions) generate top-down predictions; prediction errors $\varepsilon_l = r_l - \hat{r}_l$ drive both inference dynamics (updating representations r_l) and weight learning (weight rate $\eta_W = 10^{-4}$). As with STDP, only convolutional weights are updated via PC; the fully connected readout layers are trained with backpropagation.

STDP (Bi and Poo, 1998) converts activations to Poisson spike trains ($T = 10$ timesteps) and uses first-spike timing to update convolutional weights:

$$\Delta w = \begin{cases} A_+ e^{-\Delta t / \tau_+} & \Delta t > 0 \text{ (LTP)} \\ -A_- e^{\Delta t / \tau_-} & \Delta t < 0 \text{ (LTD)} \end{cases} \quad (3)$$

with $\tau_+ = \tau_- = 20$ ms, $A_+ = A_- = 0.003$, $\text{lr} = 5 \times 10^{-4}$. Only the convolutional layers (Conv1–Conv3) are updated via STDP; the fully connected layers (FC1, FC2) are trained with backpropagation to provide a supervised readout, analogous to a linear probe on STDP-learned features. The ~~63% CIFAR-10 accuracy~~ STDP test accuracy (57.4%) therefore reflects STDP feature quality evaluated through a BP-trained classifier, not end-to-end STDP learning.

3.4 RSA Pipeline

Feature extraction. For each layer, activations were extracted on the 720 THINGS stimuli (resized to 224×224 , standard ResNet input resolution). Convolutional activations were spatially averaged (global average pooling). All results are averaged across 5 random seeds per learning rule; ~~error bars in figures reflect seed variability (mean $\pm$ std).~~

RDM construction. Model RDMs were computed as $1 - r$ (correlation distance).

~~Brain RDMs were averaged across subjects to produce a single mean-brain RDM per ROI.~~

RSA score. Model-brain alignment ~~was quantified as~~ is Spearman’s ρ between the upper triangles of ~~model and mean-brain RDMs~~ a model RDM and a subject RDM, restricted to the stimulus pairs defined below. The main convention (ρ_{subj}) correlates the model RDM with each subject’s RDM and averages over the three subjects. The convention of earlier versions (ρ_{mean}) correlates the model RDM with the average of the three subject RDMs; it is reported alongside.

~~Bootstrap confidence intervals~~ Stimulus pairs (cross-run). 95% CIs were estimated by bootstrap resampling ($N = 10,000$) of the stimulus pairs from the mean-brain RDM comparison. ~~This sample size is sufficient for stable CI estimation at the correlation values observed here. Each subject saw the 720 stimuli in ten runs, and sub-01 and sub-02 saw them in identical order. Responses within a run can share run-specific components (baseline drifts, run-wise noise), so for two subjects with the same order the dissimilarities of same-run pairs share a component that has nothing to do with the stimuli. This inflated the agreement between sub-01 and sub-02 (at LOC 0.062 on all pairs vs. 0.023 on cross-run pairs, while sub-01 vs. sub-03 is 0.008 and 0.009) and thereby the lower bound at LOC and IT (appendix, Table 5). We therefore use only stimulus pairs that lie in different runs for all three subjects: 210,205 of 258,840 pairs; 18.8% are excluded. Within a run, dissimilarity also depends on the distance between trials (Spearman correlation between dissimilarity and lag -0.30 to -0.10); at LOC and IT, neighbouring trials are on average anticorrelated (mean dissimilarity at lag 1 up to 1.030). This would be consistent with collinearity between the overlapping hemodynamic responses of neighbouring trials in single-trial estimates,~~

which is a further reason to exclude these pairs; we did not test this explanation. Excluding same-run pairs follows the same logic as run-wise cross-validation of dissimilarities, which keeps noise shared within a run out of the dissimilarity estimates (Walther et al., 2016).

Noise ceiling. ~~Upper and lower bounds were estimated using split-half reliability corrected by the Spearman-Brown formula (Kriegeskorte et al., 2008). The lower bound follows Nili et al. (2014): each subject’s RDM is correlated with the average RDM of the other two subjects, and the three correlations are averaged. It estimates how well a model that captures the structure shared across subjects would score, and is used here as a floor: a model below it leaves reliable shared structure unexplained. The upper bound (correlation of each subject with the average of all three, including itself) is not informative with three subjects: under a permutation null in which stimulus labels are shuffled independently per subject (1000 permutations), the upper bound stays large (V1: 0.548 against an observed 0.572), because each subject contributes a third of the average it is compared with. It is therefore not shown in the figures. Under the same null the lower bound is 0.000 ± 0.002 (V1), so the observed lower bounds are far from chance at every ROI.~~

3.5 Statistical Tests

~~Pairwise differences between conditions were assessed using permutation tests ($N = 1,000$). For each pair of conditions A and B , we computed the observed difference $\Delta\rho = \rho(A, \text{brain}) - \rho(B, \text{brain})$. Under $H_0: \Delta\rho = 0$, the brain RDM vector was randomly permuted to generate a null distribution of $\Delta\rho$ values; p is the fraction of null $|\Delta\rho|$ values exceeding the observed $|\Delta\rho|$. Because the minimum resolvable p -value at $N = 1,000$ is $p = 0.001$, all p -values reported as $p < 0.001$ should be interpreted as $p \leq 0.001$.~~ **Uncertainty.** 95% confidence intervals come from a stimulus bootstrap: 1000 resamples of the 720 stimuli with replacement (seed 20261002), shared by all conditions, ROIs and quantities. Within each resample, pairs of a stimulus with itself are dropped and the cross-run pair set is recomputed. Differences between conditions, and between a condition and the lower bound, are computed on the same resamples, so their intervals account for the correlation between them. We call a difference significant when its percentile interval excludes 0. The three statements the conclusions rest on are treated as primary comparisons: the untrained network vs. BP at V1 and V2, BP vs. the untrained network at LOC, and FA vs. each other condition at V1 (7 of the 40 pairwise comparisons ~~10 pairs $\times$ 4 ROIs~~ were corrected for multiple comparisons using the). They were designated in this revision, not before the original analysis. All other pairwise comparisons are exploratory. No correction for multiple testing is applied in the main text; all 40 intervals are reported, with the primary ones marked (appendix, Table 10),

and Benjamini–Hochberg FDR procedure. Effect sizes are reported as descriptive Cohen’s d computed from per-subject RSA score distributions ($N = 3$; these values should be interpreted with caution given the small sample and Holm corrections of the primary family are given in the appendix (Table 7).

3.5 Partial RSA

To control for low-level pixel similarity, we computed a pixel RDM is computed as the pairwise correlation distance between flattened 224×224 RGB images the flattened RGB images (resized to 224px on the shorter edge and centre-cropped to 224×224 , as for the networks). Partial Spearman ρ was is estimated by residualizing both model and the ranks of the model and the brain RDM vectors against the pixel RDM via rank-based linear regression, on the ranks of the pixel RDM vector and correlating the residuals. The pixel RDM was rebuilt for this version from the definition in the code of earlier versions; applied to the earlier analysis it reproduces the stored values to within 0.00005 (they were stored at four decimals). Partial ρ is computed per seed and subject on cross-run pairs and averaged as ρ_{subj} , with the same stimulus bootstrap.

4 Results

4.1 Task Performance

Table 1: CIFAR-10 test accuracy (10,000 test images) of the repaired models after 40 epochs. Seed 0 only; checkpoints of the other four seeds were not stored, so no standard deviation can be given. Earlier versions reported training accuracy (appendix, Table 11).

Condition Accuracy (%) Backpropagation 82.4 STDP 63.2 Predictive Coding 56.6 Feedback Alignment 39.0 Random Weights 10.0

Condition	Test accuracy (%)
Backprop	75.0
STDP	57.4
Predictive Coding	56.7
Feedback Alignment	38.8
Random Weights	10.0

4.2 Layer-wise Brain Alignment

Interpreting absolute RSA scores. Absolute Spearman ρ values in this study are low (e.g., $\rho < 0.10$ at V1), which may appear surprising in light of Brain-Score results where encoding models reach $\rho \approx 0.5$ for V1. However, they are not comparable with the scores of encoding-model benchmarks: RSA and encoding models

measure fundamentally different quantities: RSA evaluates the geometry of the representational space (pairwise dissimilarity structure), whereas encoding models predict individual voxel responses, which is a more permissive criterion that can exploit stimulus-specific variance. RSA scores are further attenuated by the spatial resolution of fMRI relative to electrophysiology, and by our relatively small CNN architecture. Critically, all conditions lie within or near the noise ceiling (upper bounds: The data themselves are of limited reliability: the lower bound is 0.051 [0.039, 0.064] at V1: -0.11, -0.024 at V2: -0.09, LOC: 0.06, IT: 0.07; lower bounds: -0.07, -0.05, -0.03, -0.04), confirming that our conditions exhaust most of the available signal given the fMRI noise level. Typical RSA Spearman ρ values in the literature range from 0.01 to 0.15 for model-fMRI comparisons at this scale (Kriegeskorte et al., 2008; Schrimpf et al., 2020) 0.018 at LOC and 0.021 at IT. Taken as a floor, it separates the conditions (model – lower bound, with 95% CI). At V1, only the untrained network reaches it (+0.002 [-0.014, 0.017]); every trained rule lies below it. At V2, the untrained network (+0.005 [-0.007, 0.017]) and STDP (-0.009 [-0.021, 0.002]) are not distinguishable from it, BP lies just below it (its interval ends less than 10^{-4} below zero), and FA and PC lie below it. At LOC and IT every condition lies below it. Reaching the lower bound is a weak criterion at V1: a single luminance value per image reaches $\rho_{\text{subj}} = 0.054$ against the same V1 data, as much as the untrained network (0.053; arXiv:2608.12408).

Table 2 provides the full numerical results. Several key patterns emerge.

V1/V2—Architecture dominates, untrained V2—Untrained exceeds BP at 224 px evaluation. In early visual cortex, the untrained random-weights baseline exceeds BP at V1 ($\rho = 0.076$ $\rho = 0.053$ vs. $\rho = 0.034$; $\Delta\rho = +0.044$, $p < 0.001$). STDP achieves the highest V1 score among trained rules ($\rho = 0.064$), followed by PC ($\rho = 0.056$). All pairwise comparisons at V1 are FDR-significant ($p < 0.001$) except PC $\rho = 0.022$; $\Delta\rho = +0.031$ [0.019, 0.043] and at V2 (0.029 vs. STDP ($p = 0.004$ 0.012; $\Delta\rho = +0.017$ [0.007, 0.026]). The same architecture-dominance pattern holds for order at V1 is Random > STDP > BP > PC > FA, and all ten pairwise differences at V1 are significant in the per-subject convention; in the mean-RDM convention BP and STDP do not differ ([-0.0113, 0.0005]), and at V2 (Random : $\rho = 0.043$, BP : $\rho = 0.019$; $\Delta\rho = +0.024$, $p < 0.001$). These results demonstrate that they do not differ in either convention. At this evaluation resolution, the convolutional architecture—local connectivity, ReLU, and pooling—is thus the primary driver of early visual alignment. When the models are evaluated at their 32 px training resolution, BP and the untrained network do not differ at V1 ($\rho_{\text{subj}} = 0.046$ vs. 0.045; $\Delta\rho = -0.0007$, and that

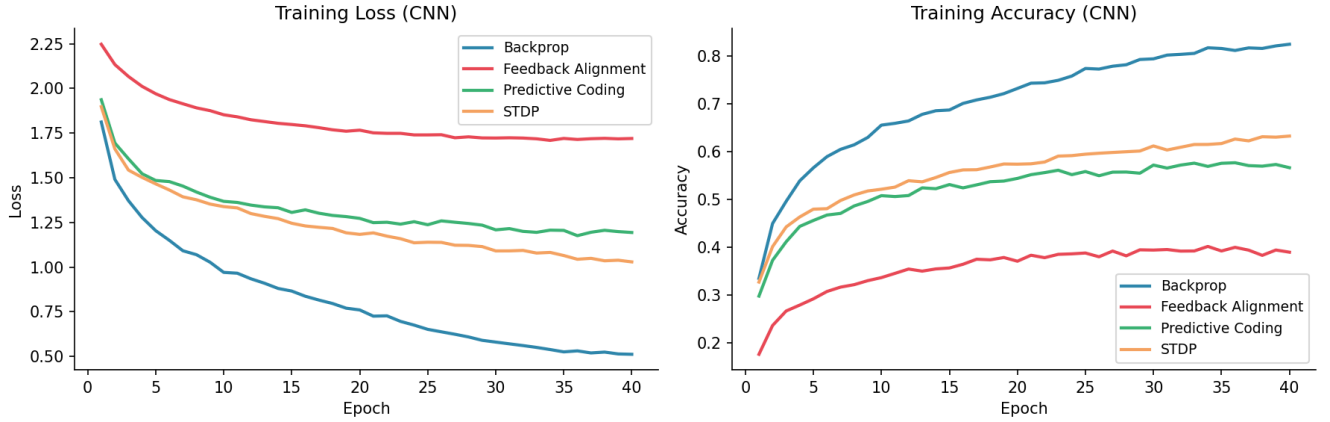


Figure 2: Training dynamics — ~~Loss of the original runs: loss~~ (left) and accuracy (right) ~~on the training batches~~ over 40 epochs ~~on CIFAR-10 (8,000 CIFAR-10 samples)~~ for all four learning rules. ~~BP converges to 82% accuracy; STDP and PC reach 63% and 57%, respectively; FA plateaus at 39%. All runs are stable without divergence. Test accuracies of the repaired models are in Table 1.~~

~~backpropagation training on a classification objective actively moves representations away from V1-like structure~~ 95% CI $[-0.0109, +0.0090]$; cross-run pairs, model set of the resolution study, same architecture and training, 5 seeds); the untrained network’s advantage at 224px arises from the difference between training and evaluation resolution (arXiv:2608.12408).

LOC Only — BP reliably exceeds random ~~LOC—Learning effect without differences among rules.~~ At LOC, BP ~~is the sole condition that reliably exceeds the~~ (0.008), PC (0.004) and STDP (0.004) exceed the random baseline ($p = 0.012$ vs. -0.003 ; BP — Random $+0.011$ $[0.002, 0.021]$, PC — Random $+0.007$ $[0.001, 0.014]$, STDP — Random $+0.007$ $[0.001, 0.014]$). FA is borderline: its difference from Random is $+0.0066$ $[-0.0002, 0.0140]$ in the per-subject convention and excludes 0 in the mean-RDM convention $[0.002, 0.026]$. No difference between two trained rules is significant (e.g. -0.005 ; $\Delta p = +0.017$, $p < 0.001$). Differences between BP and the other trained rules (FA, PC, STDP) do not survive FDR correction ($p = 0.051$ – 0.079), indicating that the LOC learning effect is modest and specific to BP BP — FA $+0.004$ $[-0.001, 0.010]$.

IT—All rules converge ~~IT—Not distinguishable at low reliability.~~ At IT, all five conditions produce similar alignment ($p = 0.008$ – 0.014). No pairwise comparison among trained rules reaches FDR significance ($p > 0.05$ for all). Even the five conditions range from 0.004 to 0.009, and none of the ten pairwise differences is significant, including Random vs. BP does not survive FDR correction at IT ($p = 0.051$). This convergence indicates that at the level of abstract categorical representations, no learning rule produces systematically more brain-like structure than any other—including no training at all.

PC and STDP lead among trained rules at V1/V2. Among trained rules, PC ($p = 0.056$) and STDP ($p = 0.064$) outperform BP ($p = 0.034$) at V1 (both $p < 0.001$). Both rules use local or spike-based updates without global error signals, yet produce more V1-like representations than BP. However, this advantage disappears at LOC and IT, where all rules converge (-0.005 $[-0.014, 0.004]$). Reliability at IT is low, second lowest after LOC (lower bound 0.021 $[0.014, 0.028]$), and every condition lies below this bound. The conditions are therefore not distinguishable at the reliability available at IT; this does not show that they are equivalent.

FA consistently lowest at V1. FA produces the lowest brain alignment at V1, V2, and LOC. At (0.009), and its differences from every other condition at V1, FA ($p = 0.012$) is significantly worse than all other conditions ($p < 0.001$ vs. Random, BP, PC, STDP), are significant in both conventions, suggesting that random feedback filters actively disrupt the representational structure that the convolutional architecture provides for free. This holds despite FA achieving 39% task-reaching 38.8% test accuracy—substantially above chance—indicating that FA learns task-relevant features that do not correspond to human visual representations. At V2, the ordering among FA, BP and PC depends on the layer assignment (Conv1 vs. Conv2; Appendix Table 6), whereas Random $>$ BP holds under both.

4.3 Hierarchical Gradient

All five conditions exhibit a sharp decline in alignment from Conv1/Conv2 to Conv3. At FC1, all conditions recover to similar levels ($p = 0.008$ – 0.014), with no rule clearly dominating are lower at LOC than at V1 (Figure 4). The random baseline shows the most distinctive pattern: highest at Conv1/Conv2 V1/V2 (ex-

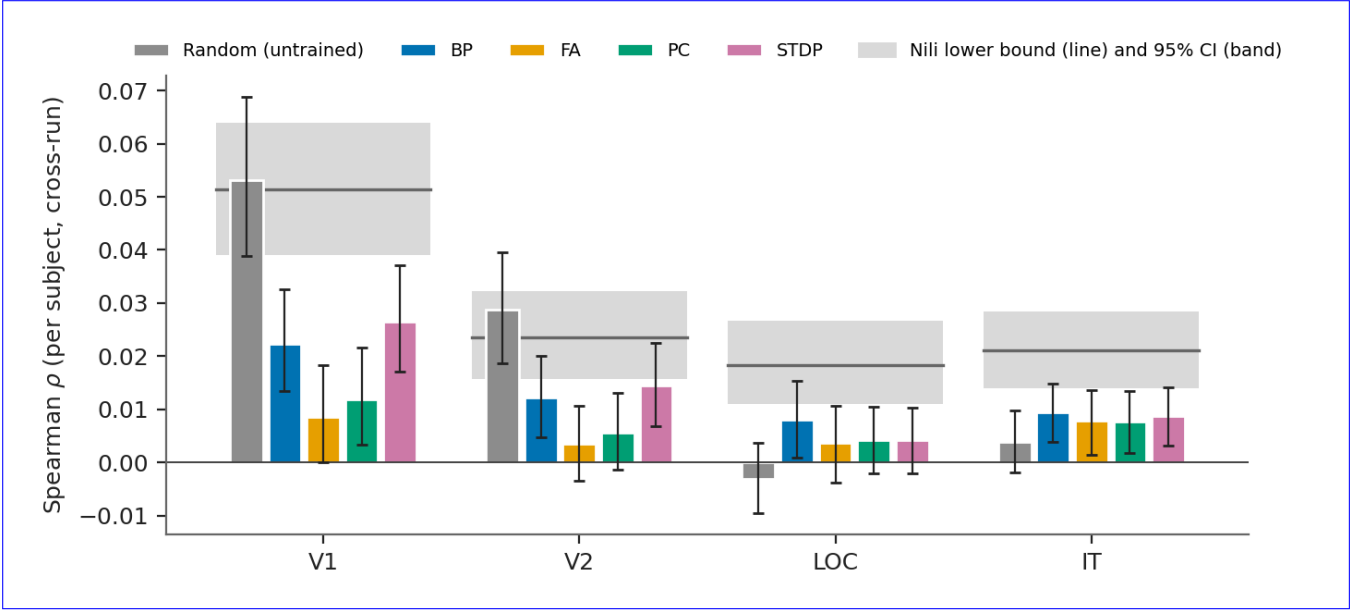


Figure 3: Brain alignment across ROIs (main result). Spearman ρ between model RDMs and ~~mean fMRI RDMs~~ (each subject’s RDM, averaged over the 3 subjects (ρ_{subj}) for each condition and ROI, cross-run stimulus pairs, mean of 5 seeds). Error bars: ~~bootstrap stimulus-bootstrap~~ 95% CI ($N=10,000$). Grey ~~hatched line and band~~: noise ceiling Nili et al. (2014) lower bound and its 95% CI. The upper bound is not shown because it is not informative with three subjects (Spearman-Brown corrected split-half Methods); white bar with black outline: untrained random-weights baseline.

Table 2: RSA scores ρ_{subj} (ρ , per-subject convention; mean across 5 seeds) with ~~stimulus-bootstrap~~ 95% ~~bootstrap~~ CIs and ~~FDR-corrected significance vs CI~~, cross-run stimulus pairs, repaired model set, 224×224 input. ~~Random Weights~~ The mean-RDM convention is in the appendix (permutation test, $N=1,000$ Table 9). Lower bound: Nili et al. (2014) leave-one-subject-out. Bold: ~~best-performing condition highest~~ ρ_{subj} per ROI (V1, V2, LOC). Input[†]: 224×224 difference from Random Weights significant (CI excludes 0). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ^{ns} $p \geq 0.05$ [‡]: at IT no pairwise difference is significant; no condition is marked.

ROI	Layer	lower bound	Random	BP	FA	PC	STDP
V1	Conv1	0.051 [.039, .064]	0.053 [.039, .069]	0.022 [.013, .033] [†]	0.009 [.000, .018] [†]	0.012 [.003, .022] [†]	0.027 [.017, .037] [†]
V2	Conv1	0.024 [.016, .032]	0.029 [.019, .040]	0.012 [.005, .020] [†]	0.004 [−.003, .011] [†]	0.006 [−.001, .013] [†]	0.015 [.007, .023] [†]
LOC	Conv3	0.018 [.011, .027]	−0.003 [−.010, .004]	0.008 [.001, .015] [†]	0.004 [−.004, .011]	0.004 [−.002, .011] [†]	0.004 [−.002, .010] [†]
IT	FC1	0.021 [.014, .028]	0.004 [−.002, .010]	0.009 [.004, .015]	0.008 [.001, .014]	0.008 [.002, .013]	0.009 [.003, .014] [‡]

ceeding all trained rules), dropping sharply at Conv3 ($\rho = -0.005$ below zero at LOC (-0.003 , the lowest), and partially recovering at FC1 ($\rho = 0.008$). This confirms that convolutional inductive biases at IT (0.004) the lowest again but not significantly different from any trained rule. Convolutional inductive biases thus drive early alignment, while later layers converge across all learning strategies at IT the differences between conditions are below what the data can resolve.

4.4 Permutation Tests Pairwise Comparisons

Pairwise condition comparisons (permutation test). Horizontal bars show $\Delta\rho = \rho_A - \rho_B$ for all ten pairwise comparisons per ROI. Green bars:

statistically significant ($p < 0.05$, FDR-corrected Benjamini-Hochberg); red bars: not significant. At V1/V2, all pairwise comparisons reach significance ($p \leq 0.014$). At LOC, only Random vs. trained rules are significant; comparisons among trained rules are non-significant after FDR correction. At IT, no comparison survives FDR correction.

Across All 40 FDR-corrected pairwise comparisons, 30 reach significance pairwise differences (10 pairs $\times$ 4 ROIs) with their bootstrap intervals are listed in Table 10. At V1/V2, all comparisons are significant ($p \leq 0.014$), confirming that the five conditions produce genuinely different early visual representations. At LOC, only comparisons involving Random Weights vs. any trained rule are significant; pairwise comparisons among trained rules (BP vs. FA, all ten are significant; at V2,

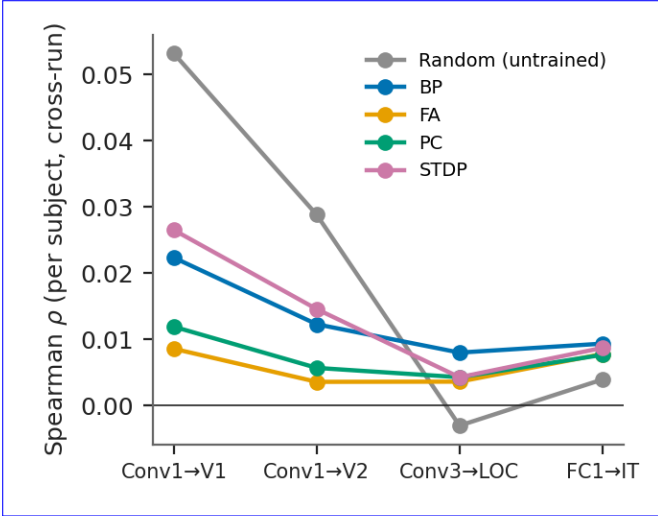


Figure 4: Hierarchical gradient of brain alignment. For each model layer, along the best-matching ROIs Spearman ρ is plotted fixed layer-to-ROI mapping (Conv1→V1, Conv1→V2, Conv3→LOC, FC1→IT). All conditions exhibit a sharp drop from Conv2 to Conv3 are lower at LOC than at V1. At FC1, all conditions recover to similar levels ($\rho = 0.008-0.014$) with no rule clearly dominating, consistent with the IT convergence finding the conditions lie between 0.004 and 0.009 and are not distinguishable. The random baseline drops most sharply untrained network is the highest at Conv3 ($\rho = -0.005$) before partially recovering V1/V2 and the lowest at FC1 LOC and IT.

all except BP vs. PC, BP vs. STDP) do not survive FDR correction ($p = 0.051-0.079$). At IT, no pairwise comparison survives FDR correction, including STDP. At LOC, the differences between the random baseline and BP, PC and STDP are significant, Random vs. BP ($p = 0.051$), confirming full convergence at the highest visual area. FA is borderline, and none of the six differences among trained rules is significant. At IT, none of the ten is significant.

4.5 Subject-Level Consistency

We assessed whether learning-rule rankings are stable across the results hold within individual subjects (Figure 5; single-subject ρ with stimulus-bootstrap CIs on the same resamples). Sub-03 shows consistently near-zero weaker alignment across all conditions and ROIs, suggesting lower fMRI signal quality. For At V1, sub-01 and sub-02, the architecture dominance pattern is consistent: Random and STDP produce the highest V1 alignment, while FA produces the lowest each show the group order Random > STDP > BP > PC > FA in their point estimates, and in each of them the untrained network exceeds every trained rule (e.g.

Random - BP: sub-01 +0.033 [0.016, 0.051], sub-02 +0.052 [0.026, 0.076]). FA is below every other condition in sub-01; in sub-02 its difference from PC is borderline. At LOC, BP leads for both subjects. Results were qualitatively unchanged when single subjects do not resolve the differences: BP has the highest point estimate in sub-02 and sub-03 was excluded (see Figure 5 for per-subject scores; sub-01 and, and in sub-01 STDP (0.007) is marginally above BP (0.006); no difference between two trained rules is significant in any subject, and the difference between the untrained network and BP is significant only in sub-02 independently replicate the ranking Random > STDP > PC > BP > FA at V1). (Random - BP -0.024 [-0.047, -0.001]). The LOC result of the main analysis therefore rests on the combination of subjects, not on a consistent per-subject ranking.

4.6 Seed Variability

To assess robustness to random initialization, we trained each learning rule with 5 different random seeds and report mean $\pm$ std RSA scores (Figure 6). At V1/V2, seed variability the seed standard deviation (0.001-0.006) is small relative to between-rule differences (std $\approx 0.003-0.007$ vs. the differences between conditions (e.g. $\Delta\rho \approx 0.044$ $\Delta\rho = +0.031$ between Random and BP), confirming that rankings are stable at V1), so the order of conditions is stable across initializations. At IT, seed variability is comparable to between-rule differences, consistent with the convergence finding that no rule reliably outperforms at higher areas the seed standard deviations are also small (0.0005-0.0020), and most seed ranges overlap; BP's lowest seed (0.00865) lies just above FA's highest (0.00862) and above the untrained network's highest (0.00725). The seed spread does not include the uncertainty from the sampling of stimuli, which is much larger (Random - BP at IT: [-0.014, 0.004]); it is this uncertainty that leaves the IT comparisons unresolved.

4.7 Best-Layer-per-ROI Analysis

We evaluated whether the fixed layer-to-ROI mapping is optimal by computing RSA for all layer $\times$ ROI combinations (Table 3; point estimates, ρ_{subj} , cross-run). For BP, the fixed mapping matches the best layer at all four ROIs. For STDP, Conv1 is optimal at V1 (matching the fixed mapping). However, for several conditions (random, FA, PC) 3 of four ROIs; at LOC, FC1 is higher than Conv3. For Random, FA and PC, FC1 produces the highest V1 alignment likely alignment (Random: 0.102 vs. 0.053 at Conv1) likely reflecting that FC1 compresses all information into a 512-dimensional space that happens to correlate with the fMRI signal. This finding suggests that best-layer analysis can be misleading when

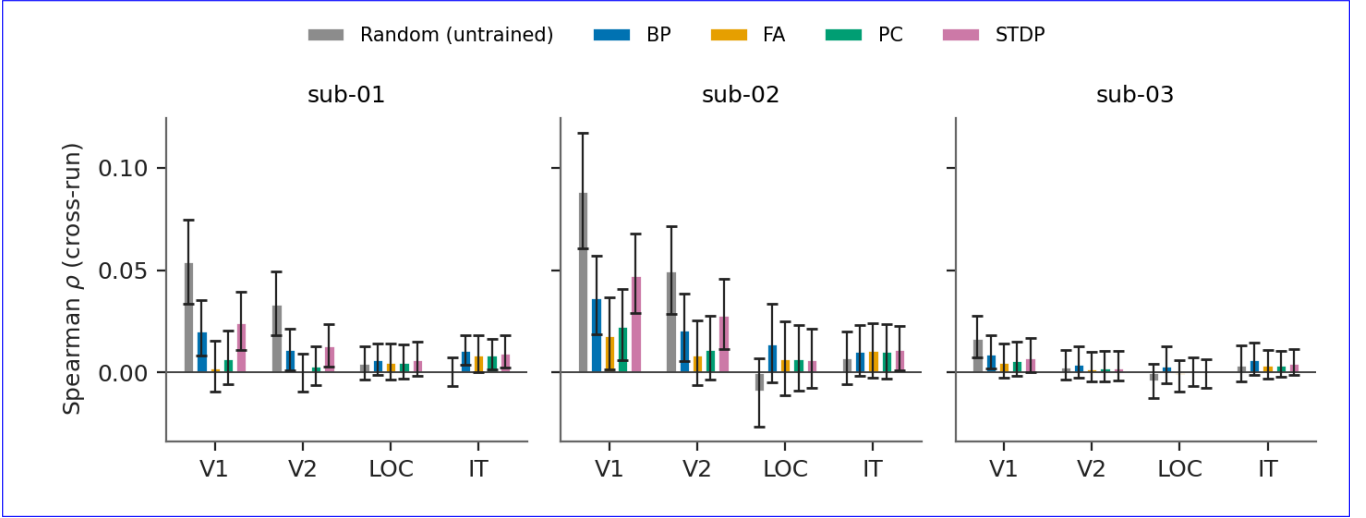


Figure 5: ~~Subject-level~~ Single-subject RSA scores per learning rule and ROI (~~sub-01: blue~~ cross-run pairs, ~~sub-02: orange~~, ~~sub-03: green~~ mean of 5 seeds), ~~with stimulus-bootstrap 95% CIs~~. Sub-03 shows ~~consistently near-zero~~ weaker alignment than sub-01 and sub-02 across ~~all conditions and ROIs, suggesting lower fMRI signal quality for this participant~~. Sub-01 ~~In sub-01 and sub-02 show consistent ranking patterns, with Random and STDP leading at the V1 /V2, and BP leading order of the point estimates is that of the group; at LOC, no difference between two trained rules is significant in any single subject.~~

layer dimensionality varies, and supports the use of fixed anatomically-motivated mappings.

Table 3: Best layer per ROI and its ρ_{subj} (cross-run, mean of 5 seeds; point estimates). In parentheses: the fixed layer’s value where it differs.

Condition	V1	V2	LOC	IT
Random	FC1 0.102 (0.053)	FC1 0.061 (0.029)	Conv1 -0.002 (-0.003)	FC1 0.004
BP	Conv1 0.022	Conv1 0.012	FC1 0.012 (0.008)	FC1 0.009
FA	FC1 0.020 (0.009)	FC1 0.014 (0.004)	FC1 0.005 (0.004)	FC1 0.008
PC	FC1 0.015 (0.012)	FC1 0.013 (0.006)	Conv2 0.005 (0.004)	FC1 0.008
STDP	Conv2 0.027 (0.027)	FC1 0.018 (0.015)	FC1 0.007 (0.004)	FC1 0.009

4.8 Partial RSA

~~After removing~~ Removing the variance explained by pixel-level similarity ~~, all conditions retain their alignment patterns~~ lowers every condition’s V1 score (Δ between -0.005 and -0.002), and each of these decreases is significant (Table 4). The ordering order at V1 is unchanged (Random > STDP > PC > BP > FA > PC > FA), and the untrained network still exceeds BP at V1 ~~is fully preserved, with small and comparable decreases across conditions ($\Delta \approx -0.004$ to -0.008). This confirms that the architecture-dominance effect reflects genuine representational structure beyond low-level image statistics ($+0.034$ [0.021, 0.046]) and V2 ($+0.018$ [0.008, 0.027]). Across all 40 pairwise comparisons, the same differences are significant with and without pixel control, so none of the results reported above is carried by pixel-level similarity.~~

~~Partial RSA controlling for pixel similarity. Left: standard RSA scores. Right: partial Spearman ρ after regressing out the pixel-similarity RDM. The rank ordering of conditions is preserved after pixel control, and the decrease in ρ is small and comparable across conditions ($\Delta \approx -0.005$ to -0.008 at V1).~~

4.9 Filter Analysis

Conv1 filters from all five conditions exhibit color-opponent patch structure rather than oriented Gabor-like patterns, consistent with CIFAR-10 training. The Gabor-peakedness score (FFT spectral peak-to-mean ratio) is highest for BP (2.00 ± 0.52 1.99 ± 0.52), followed by FA (1.82 ± 0.36 1.82 ± 0.36), STDP (1.79 ± 0.47 1.79 ± 0.47), PC (1.73 ± 0.45 1.73 ± 0.45), and Random (1.73 ± 0.45 1.73 ± 0.45). The differences are small relative to within-rule variability and standard deviations overlap substantially across all conditions, indicating that no learning rule produces markedly more Gabor-like filters than random initialization at this scale.

5 Discussion

Architecture as the primary driver of early alignment. Our most striking finding is that an untrained CNN *exceeds* backpropagation at V1 ($\rho = 0.076$ $\rho = 0.053$ vs. 0.034 , $\Delta\rho = +0.044$, $p < 0.001$ 0.022 , $\Delta\rho = +0.031$ [0.019, 0.043]). This goes beyond the observation of Saxe et al. (2011) that random networks

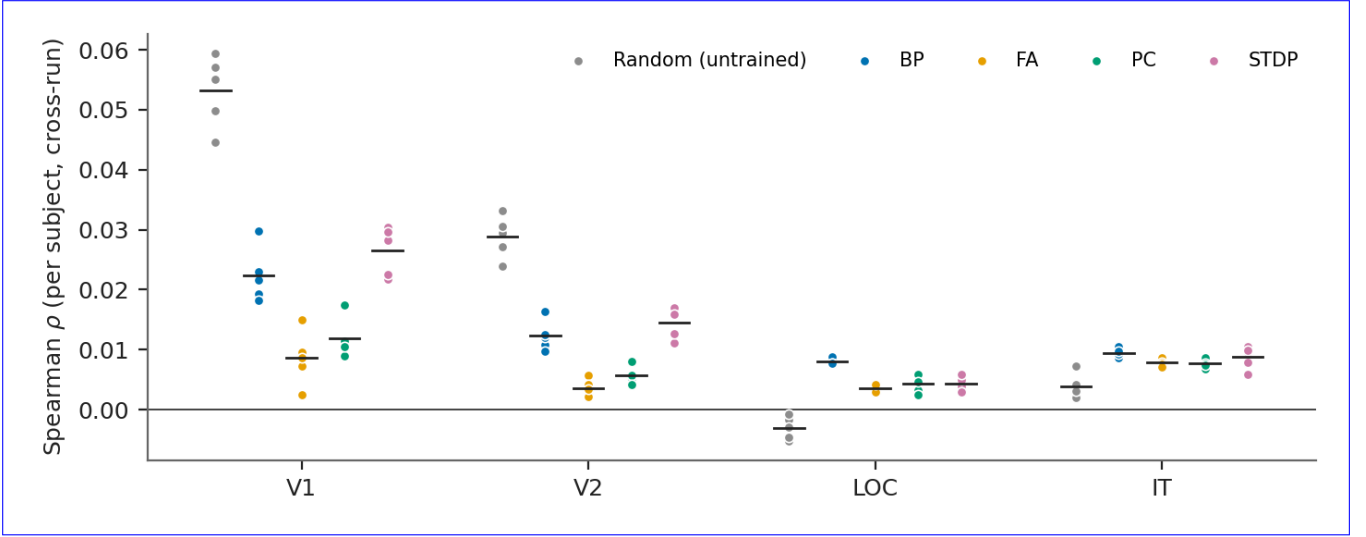


Figure 6: RSA scores (mean $\pm$ std across of the 5 random seeds, 224×224 input per condition (dots) and their mean (line), ρ_{subj} , cross-run pairs. Error bars reflect variability across initializations. At V1/V2, between-rule differences are large relative to within-rule the seed variability, confirming that rankings are stable across initializations spread. At IT, the seed variability overlaps across rules spread is small, consistent with and the convergence finding lack of significant differences reflects the stimulus-sampling uncertainty.

Table 4: Standard vs. (ρ_{subj}) and partial RSA at V1 (ρ_{partial} , pixel similarity controlled, 224×224) at V1 and V2; $\Delta = \rho_{\text{partial}} - \rho_{\text{subj}}$. Partial RSA was computed from a single seed averaged Cross-run pairs, repaired model RDM rather than averaging partial ρ across seeds set, which accounts for minor differences from Table 2 (e.g. per-subject convention, Random $\rho_{\text{std}} = 0.078$ here vs mean of 5 seeds, stimulus-bootstrap 95% CI: 0.076 in Table 2).

Condition ρ_{std} ρ_{partial} Δ Random 0.078 0.074 -0.004 STDP 0.067 0.061 -0.005 PC 0.058 0.054 -0.004 BP 0.034 0.026 -0.008 FA 0.012 0.005 -0.007

Condition	V1			V2		
	ρ_{subj}	ρ_{partial} [95% CI]	Δ [95% CI]	ρ_{subj}	ρ_{partial} [95% CI]	Δ [95% CI]
Random	0.053	0.051 [0.037, 0.066]	-0.0025 [-0.0042, -0.0011]	0.029	0.028 [0.018, 0.038]	-0.0011 [-0.0024, -0.0000]
STDP	0.027	0.022 [0.013, 0.032]	-0.0045 [-0.0069, -0.0024]	0.015	0.013 [0.005, 0.020]	-0.0020 [-0.0041, -0.0002]
BP	0.022	0.017 [0.009, 0.027]	-0.0052 [-0.0080, -0.0027]	0.012	0.010 [0.003, 0.017]	-0.0022 [-0.0045, -0.0002]
PC	0.012	0.007 [-0.001, 0.017]	-0.0046 [-0.0071, -0.0026]	0.006	0.004 [-0.003, 0.011]	-0.0021 [-0.0041, -0.0005]
FA	0.009	0.004 [-0.004, 0.013]	-0.0045 [-0.0069, -0.0024]	0.004	0.002 [-0.005, 0.008]	-0.0020 [-0.0040, -0.0003]

have non-trivial visual structure: in our setup, at 224 px evaluation, the convolutional inductive biases (local connectivity, ReLU, pooling) are not merely comparable to BP but produce higher early visual alignment. This suggests that, at least for small CNNs trained on CIFAR-10, backpropagation on a classification objective moves. The advantage is not evidence that training degrades early-layer representations away from V1-like structure. Whether this structure: when the models are evaluated at their 32 px training resolution, BP and the untrained network do not differ at V1 ($\rho_{\text{subj}} = 0.046$ vs. 0.045; $\Delta\rho = -0.0007$, 95% CI [-0.0109, +0.0090], model set of the resolution study), and the untrained network's lead at 224 px arises from the difference between training and evaluation resolution (arXiv:2608.12408). Whether the same holds at larger scales (e.g., ResNet-50 on ImageNet) remains to be tested.

Among trained rules, STDP ($\rho = 0.064$) and PC ($\rho = 0.056$) preserve more V1-like structure than BP, consistent with the idea that local unsupervised learning retains more of the architectural inductive bias (Olshausen and Field, 1996).

On the untrained baseline. The untrained condition's batch-normalisation layers hold their initialised statistics, so in evaluation mode they act as the identity, whereas every trained condition carries statistics accumulated during training. A 2×2 calibration control varying the calibration set and the calibration resolution, with the convolutional weights held bit-identical, indicates that this does not produce the result: with statistics calibrated at the evaluation resolution the untrained network still exceeds backpropagation at V1 by +0.041 (CIFAR-10 statistics) or +0.033 (statistics from a disjoint pool of THINGS images), 5/5 and 5/5 seeds (original analysis: all

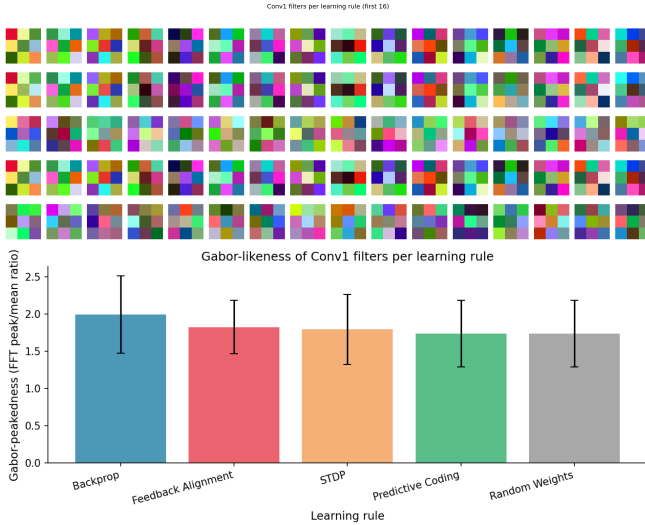


Figure 7: Conv1 filter analysis for all five conditions (original model set; weight-based, so not affected by the evaluation defect). (a) First 16 Conv1 filters per learning rule. All conditions show color-opponent patch structure rather than oriented Gabor patterns, consistent with CIFAR-10 training at $224 \times 224 \times 32$. (b) Gabor-peakedness (FFT spectral peak-to-mean ratio): BP 2.00 ± 0.52 , FA 1.99 ± 0.52 , FA 1.82 ± 0.36 , STDP 1.79 ± 0.47 , PC 1.73 ± 0.45 , Random 1.73 ± 0.45 . Differences are small relative to within-rule variability.

pairs, mean-RDM convention). The result is, however, strongly dependent on the evaluation resolution, which this paper holds fixed at 224 px; see arXiv:2608.12408.

Learning rules converge Few reliable differences at higher areas. At LOC, only BP significantly exceeds BP. PC and STDP exceed the random baseline, but FA is borderline, and the differences among trained rules are small and non-significant after FDR correction not significant. At IT, all five conditions converge completely. This convergence no two conditions can be distinguished. This may reflect a genuine property of abstract categorical representations, but is also consistent with a capacity limitation: our small CNN trained on 8,000 CIFAR-10 samples may simply lack the representational power to differentiate at higher layers. Prior work with larger models suggests that BP-trained networks have a systematic IT advantage (Schrimpf et al., 2020); whether this advantage reflects the learning rule or the additional model capacity remains an open question. We note that the noise ceiling at IT is also the lowest of all ROIs (upper bound: 0.07), leaving limited room for any condition to differentiate — it is possible that true differences. The low reliability of the IT data (lower bound 0.021, with every condition below it) also leaves little room for differences to be resolved with three subjects; true differences may exist but are unresolvable

given the fMRI signal quality at this ROI and sample size ($N = 3$) not resolvable here.

PC and STDP as biologically plausible routes to FA lowest at V1 alignment. PC ($p = 0.056$) and STDP ($p = 0.064$) outperform BP ($p = 0.034$) at V1 — the cortical area where biological plausibility is most constrained by known physiology. Both rules use local updates without global error signals, suggesting that local computation preserves more of the architectural inductive bias that underlies V1-like representations. This is consistent with the theoretical argument that PC approximates BP under certain conditions (Whittington and Bogacz, 2017), but the superior V1 alignment of PC and STDP over BP suggests they may be more appropriate models of early visual cortex than BP.

FA consistently lowest. FA produces the lowest alignment among trained rules at V1, V2, and LOC, significantly below all other conditions at V1/V2 ($p < 0.001$), despite achieving 39% task, despite reaching 38.8% test accuracy. This is consistent with random feedback filters disrupting the representational structure that convolutional architecture provides, though we cannot rule out that this reflects an optimization artifact specific to FA in convolutional networks rather than a fundamental incompatibility. At V2 After pixel control, FA’s partial RSA score drops to near zero ($p_{\text{partial}} = 0.001$, $p = 0.755$), indicating that essentially none of FA’s V2 alignment survives pixel-similarity control V2 score is 0.002 $[-0.005, 0.008]$; it was already not distinguishable from zero without pixel control (0.004 $[-0.003, 0.011]$). This result may be specific to CNNs, where spatial structure in feature maps interacts poorly with random feedback filters (Lillicrap et al., 2016).

Dissociation between task performance and brain alignment. Task accuracy and brain alignment are dissociated throughout the hierarchy in our small-scale setting. At V1, the untrained baseline (10% 10.0% test accuracy) exceeds BP (82% accuracy 75.0%), and STDP (63%) and PC (57%) both outperform BP despite lower task performance. At IT, all rules converge regardless of accuracy at IT, conditions with test accuracies between 10.0% and 75.0% cannot be distinguished. This suggests that, at least for small CNNs trained on CIFAR-10, the relevant axis is not “how well does it classify” but “what representational structure does the objective induce at each layer.” Whether this dissociation persists at ImageNet scale, where task-optimized models achieve substantially higher Brain-Score alignment (Schrimpf et al., 2020), remains an open question.

Implications. Our results carry two concrete methodological messages. For computational neuroscience: studies comparing trained models to brain data must always include an untrained baseline with the same architecture. Without this control, architec-

ture effects are confounded with learning effects—our data show this confound can be essentially complete at V1. ~~For machine learning: the inductive biases of the convolutional architecture (local connectivity, weight sharing, ReLU, pooling) are more important than the weight-update rule for~~ The same control must be run at the training resolution: the untrained network's lead over BP at V1 is present at 224px evaluation but absent at the 32px training resolution; it arises from the mismatch between training and evaluation resolution, not from training per se degrading early-layer feature representations. ~~When designing architectures for brain-inspired AI, the structural priors deserve as much attention as the learning algorithm structure (arXiv:2608.12408).~~ A second message comes from the revision itself: when subjects share a presentation order, same-run stimulus pairs can inflate between-subject agreement and the noise ceiling, and should be excluded or modelled.

5.1 Limitations

Several limitations bound our conclusions. (1) All models were trained on CIFAR-10 using a small custom CNN, far below the scale at which ResNet-50 or ViT models achieve competitive Brain-Score results. The small architecture limits representational capacity and may ~~explain the IT convergence~~ contribute to the absence of differences at IT. (2) We have only 3 subjects, with sub-03 showing consistently weak signal; our core findings are driven by sub-01 and sub-02. With three subjects only the lower noise bound is informative. Replication on a larger dataset (e.g., Natural Scenes Dataset, $N = 8$) would strengthen generalizability. (3) STDP was applied to static images via Poisson spike trains, discarding the temporal dynamics that STDP was designed to exploit. (4) The best-layer-per-ROI analysis reveals that FC1 sometimes produces higher V1 alignment than Conv1, complicating the anatomical mapping assumption. (5) Gabor-peakedness scores show no significant differences across learning rules, but this metric captures only a coarse proxy for V1-like filter structure. (6) ~~Bootstrap CIs resample stimulus pairs rather than stimuli, which may underestimate uncertainty~~ Restricting the analysis to cross-run pairs removes 18.8% of the stimulus pairs; it removes the shared run component, but the sensitivity analyses (appendix) show that the main conclusions do not depend on this choice. (7) We use a single dataset (THINGS-fMRI); cross-dataset replication would strengthen all claims. (8) Models were trained at 32×32 resolution but evaluated at 224×224 ; ~~random filters are approximately scale-invariant while trained filters are optimized for 32×32 statistics, which may partly contribute to the random~~ the size of the untrained baseline's advantage at V1 advantage depends strongly on the evaluation resolution, which is quantified in arXiv:2608.12408.

6 Conclusion

We presented a systematic RSA comparison of four learning rules plus an untrained baseline at 224×224 resolution, averaged across 5 seeds, against the THINGS-fMRI benchmark. In our small-scale setup, the learning rule has little effect on cortical alignment: architecture dominates early visual representations (V1/V2), where the untrained baseline exceeds all trained rules ~~and, at V1, is the only condition that reaches the lower noise bound.~~ This advantage holds at the 224px evaluation resolution; in the model set of the resolution study (same architecture and training, 5 seeds) the V1 gap is $\Delta\rho = +0.0304$ (95% CI $[+0.0181, +0.0423]$) at 224px, and when the models are evaluated at their 32px training resolution it vanishes ($\Delta\rho = -0.0007$, and all rules converge at higher areas (IT) 95% CI $[-0.0109, +0.0090]$). BP and the untrained network do not differ at V1, and the untrained network's lead at 224px arises from the difference between training and evaluation resolution (arXiv:2608.12408). At LOC, ~~with no reliable pairwise differences—though this convergence may partly reflect model capacity limitations.~~ Only LOC shows a selective learning effect, where BP alone reliably exceeds the random baseline. Among trained rules, PC and STDP preserve more V1-like structure than BP, suggesting that local unsupervised learning retains architectural inductive biases ~~that global error signals may disrupt~~ three of the four trained rules exceed the untrained baseline (FA borderline) without differing from each other; at IT, no two conditions can be distinguished at the low reliability of the data—though this may also reflect model capacity limitations. FA consistently produces the lowest alignment ~~despite meaningful task accuracy at V1 despite above-chance accuracy (38.8%).~~ These results suggest that, at least at this scale, ~~the search for biologically plausible learning rules should focus on preserving architectural inductive biases~~ learning-rule comparisons at early areas, rather than optimizing for task performance are dominated by the architecture and by the relation between training and evaluation resolution, and should report both. Whether these findings generalize to larger architectures trained on richer datasets remains an important open question.

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Code Availability

Code and results: <https://github.com/nilsleut/learning-rules-rsa/tree/axiv-v4>.

A Sensitivity analyses

Pair set and block cleaning. Table 5 gives the noise bounds and the per-subject model scores for three pair sets: all stimulus pairs (as in earlier versions), cross-run pairs (main analysis), and cross-run pairs with block cleaning, in which, separately for each subject, the mean of every run-pair block is subtracted from that subject’s RDM and from the model RDM it is compared with. Block cleaning is defined only for the per-subject convention: the three subjects have different run structures, so no single cleaning applies to their average. The last column gives the original (pre-repair) model set on cross-run pairs. Restricting to cross-run pairs lowers the lower bound by 10 % at V1, 28 % at V2, 46 % at LOC and 39 % at IT; block cleaning changes it little. The model scores change by at most 0.002 between variants.

Conclusions under each variant. The statements of the main text (untrained exceeds BP at V1 and V2; BP exceeds the untrained network at LOC; FA lowest at V1; no significant differences among trained rules at LOC; none at IT) have the same sign and the same significance in all three pair sets, with one exception. In the repaired model set with block cleaning, BP vs. FA at LOC becomes borderline significant: $+0.0050$ $[0.0001, 0.0102]$, against $+0.0044$ $[-0.0011, 0.0102]$ on cross-run pairs without cleaning. We treat it as borderline.

Layer assignment of V2 (Table 6). The fixed mapping assigns V2 to Conv1. With V2 assigned to Conv2 instead (same pair set, resamples and per-subject convention), the untrained network still exceeds BP, and it and STDP still exceed FA; BP falls below FA, and PC is no longer distinguishable from FA. The ordering among FA, BP and PC at V2 therefore depends on the layer assignment, whereas $\text{Random} > \text{BP}$ holds under both.

Correction for multiple comparisons (Table 7). Two-sided bootstrap p -values from the same 1000 resamples, $p = (2 \min(\#\{d_b \leq 0\}, \#\{d_b \geq 0\}) + 1) / (B + 1)$, corrected by Benjamini-Hochberg (false discovery rate) and by Holm (family-wise error rate), each at the 5 % level, within the current family of 7 primary comparisons and within the earlier family of 11, which also contained FA against every other condition at V2. All 11 comparisons of the earlier family, including the four at V2, are significant under both corrections, and all 7 of the current family under both corrections in both families.

Diagnostics of the run structure (Table 8). If

the run component were shared beyond same-run pairs, the mean dissimilarity of each pair of runs would agree between sub-01 and sub-02. The Spearman correlation of their 45 run-pair means is small and not significant at any ROI (exact one-sided p over all 3,628,800 relabellings of sub-02’s runs). With only 45 run pairs this test has little power, so block cleaning is reported as a robustness check rather than ruled out. Block cleaning changes the V1 luminance fit by -2% for sub-03, whose run order differs, and by -1% (sub-01) and -10% (sub-02); the pre-specified stop criterion (a decrease of more than 30 % for sub-01 or sub-02) is not met.

B Errata of earlier versions

Accuracy. Table 11 gives the accuracies printed in earlier versions next to the test accuracies of this version. The earlier values are accuracies on training batches in the last epoch of the original runs, measured with batch normalisation in training mode; for PC and STDP this is also the mode in which the evaluation defect left them. They are not test accuracies and are not comparable to Table 1.

Table 12 lists values printed in v2 next to the value its own source gives. The first four are double-rounding errors (rounding first to four, then to three decimals). The V1 gap printed in v2 ($+0.044$) is that of the resolution study (arXiv:2608.12408, $+0.044$), while the table it accompanied gives $+0.042$. The noise bounds printed in v2 (upper 0.11, 0.09, 0.06, 0.07; lower 0.07, 0.05, 0.03, 0.04) could not be traced to any stored result and are replaced by the bounds above.

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Table 5: [Sensitivity: noise bounds and \$\rho_{\text{subj}}\$ \(mean of 5 seeds\) for all pairs, cross-run pairs \(main\), and cross-run pairs with per-subject block cleaning; repaired model set. Last column: original model set, cross-run pairs. Lower bound with stimulus-bootstrap 95% CI. The upper bound is listed for completeness; it is not informative at \$N = 3\$ \(Methods\).](#)

ROI	Quantity	all pairs	cross-run	block-cleaned	cross-run, original set
V1	lower bound	0.057 [0.045, 0.069]	0.051 [0.039, 0.064]	0.054 [0.041, 0.066]	–
	upper bound	0.575	0.572	0.572	–
	Random	0.053	0.053	0.053	0.053
	BP	0.021	0.022	0.021	0.024
	FA	0.007	0.009	0.006	0.009
	PC	0.011	0.012	0.010	0.039
	STDP	0.025	0.027	0.024	0.045
V2	lower bound	0.033 [0.025, 0.041]	0.024 [0.016, 0.032]	0.026 [0.018, 0.034]	–
	upper bound	0.568	0.563	0.564	–
	Random	0.029	0.029	0.028	0.029
	BP	0.011	0.012	0.010	0.013
	FA	0.003	0.004	0.002	0.004
	PC	0.005	0.006	0.004	0.019
	STDP	0.014	0.015	0.012	0.024
LOC	lower bound	0.034 [0.027, 0.042]	0.018 [0.011, 0.027]	0.019 [0.011, 0.027]	–
	upper bound	0.566	0.558	0.559	–
	Random	–0.002	–0.003	–0.005	–0.003
	BP	0.007	0.008	0.008	0.008
	FA	0.003	0.004	0.003	0.004
	PC	0.004	0.004	0.003	0.003
	STDP	0.004	0.004	0.003	0.004
IT	lower bound	0.035 [0.028, 0.042]	0.021 [0.014, 0.028]	0.021 [0.014, 0.028]	–
	upper bound	0.563	0.556	0.556	–
	Random	0.005	0.004	0.003	0.004
	BP	0.008	0.009	0.009	0.010
	FA	0.007	0.008	0.006	0.008
	PC	0.007	0.008	0.007	0.008
	STDP	0.008	0.009	0.008	0.007

Table 6: [Primary V2 differences \(\$\rho_{\text{subj}}\(A\) - \rho_{\text{subj}}\(B\)\$, stimulus-bootstrap 95% CI; cross-run pairs, repaired model set\) with V2 assigned to Conv1 \(main analysis\) or to Conv2.](#)

Difference at V2	Conv1→V2 (main)	Conv2→V2	same sign and significance
Random – BP	+0.0165 [0.0072, 0.0261]	+0.0526 [0.0401, 0.0656]	yes
Random – FA	+0.0252 [0.0150, 0.0361]	+0.0313 [0.0212, 0.0413]	yes
BP – FA	+0.0087 [0.0053, 0.0125]	–0.0213 [–0.0274, –0.0148]	no
PC – FA	+0.0021 [0.0005, 0.0037]	–0.0005 [–0.0035, 0.0027]	no
STDP – FA	+0.0110 [0.0077, 0.0146]	+0.0097 [0.0047, 0.0152]	yes

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Table 7: [All 11 comparisons of the earlier primary family with bootstrap \$p\$ -values and adjusted \$p\$ -values \(Benjamini–Hochberg, Holm\) within the current family of 7 \(above the line\) and within the earlier family of 11 \(all rows; the four below the line belonged to the earlier family only\). Per-subject convention, cross-run pairs, repaired model set. The smallest attainable bootstrap \$p\$ is \$1/\(B+1\)\$.](#)

Comparison	ROI	$\Delta\rho_{\text{subj}}$ [95% CI]	p_{boot}	BH, 7	BH, 11	Holm, 7	Holm, 11
Random – BP	V1	+0.0308 [0.0186, 0.0428]	0.0010	0.0014	0.0014	0.0070	0.0110
Random – BP	V2	+0.0165 [0.0072, 0.0261]	0.0010	0.0014	0.0014	0.0070	0.0110
Random – BP	LOC	−0.0110 [−0.0206, −0.0023]	0.0190	0.0190	0.0190	0.0190	0.0190
Random – FA	V1	+0.0446 [0.0313, 0.0580]	0.0010	0.0014	0.0014	0.0070	0.0110
BP – FA	V1	+0.0138 [0.0092, 0.0186]	0.0010	0.0014	0.0014	0.0070	0.0110
FA – PC	V1	−0.0033 [−0.0054, −0.0011]	0.0030	0.0035	0.0037	0.0070	0.0110
FA – STDP	V1	−0.0179 [−0.0229, −0.0130]	0.0010	0.0014	0.0014	0.0070	0.0110
Random – FA	V2	+0.0252 [0.0150, 0.0361]	0.0010	–	0.0014	–	0.0110
BP – FA	V2	+0.0087 [0.0053, 0.0125]	0.0010	–	0.0014	–	0.0110
FA – PC	V2	−0.0021 [−0.0037, −0.0005]	0.0090	–	0.0099	–	0.0180
FA – STDP	V2	−0.0110 [−0.0146, −0.0077]	0.0010	–	0.0014	–	0.0110

Table 8: [Diagnostics of the run structure. Top: agreement of the run-pair mean dissimilarities between sub-01 and sub-02 \(identical presentation order\). Bottom: V1 luminance fit per subject on cross-run pairs before and after block cleaning.](#)

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ROI	ρ (sub-01 vs sub-02, 45 run-pair means)	p (exact, one-sided)
V1	0.055	0.336
V2	0.065	0.323
LOC	0.171	0.160
IT	0.097	0.226
Subject	V1 luminance fit: cross-run $\rightarrow$ block-cleaned (change)	
sub-01	0.054 $\rightarrow$ 0.054 (−1 %)	
sub-02	0.096 $\rightarrow$ 0.086 (−10 %)	
sub-03	0.011 $\rightarrow$ 0.011 (−2 %)	

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Table 9: RSA scores in the mean-RDM convention ρ_{mean} (model RDM vs. the average of the three subject RDMs; the convention of earlier versions), with stimulus-bootstrap 95% CI; cross-run pairs, repaired model set, mean of 5 seeds. Bold, [†], [‡] as in Table 2. The lower bound is defined per subject and is given in Table 2.

ROI	Layer	lower bound	Random	BP	FA	PC	STDP
V1	Conv1	–	0.077 [.055, .099]	0.034 [.020, .050] [†]	0.014 [.000, .028] [†]	0.018 [.006, .034] [†]	0.039 [.025, .056] [†]
V2	Conv1	–	0.043 [.027, .061]	0.019 [.007, .031] [†]	0.005 [–.006, .017] [†]	0.009 [–.003, .020] [†]	0.022 [.010, .035] [†]
LOC	Conv3	–	–0.007 [–.018, .004]	0.013 [.001, .027] [†]	0.006 [–.006, .019] [†]	0.007 [–.004, .018] [†]	0.007 [–.004, .017] [†]
IT	FC1	–	0.006 [–.003, .016]	0.015 [.006, .024]	0.014 [.004, .024]	0.013 [.003, .023]	0.015 [.005, .024] [‡]

Table 10: All pairwise differences $\rho_{\text{subj}}(A) - \rho_{\text{subj}}(B)$ with stimulus-bootstrap 95% CI (cross-run pairs, repaired model set). Bold: CI excludes 0. ^P: primary comparison (Methods); all others are exploratory, without correction for multiple testing.

Pair (A – B)	V1	V2	LOC	IT
Random – BP	+ 0.0308 [0.0186, 0.0428] ^P	+ 0.0165 [0.0072, 0.0261] ^P	– 0.0110 [–0.0206, –0.0023] ^P	–0.0054 [–0.0144, 0.0036]
Random – FA	+ 0.0446 [0.0313, 0.0580] ^P	+ 0.0252 [0.0150, 0.0361]	–0.0066 [–0.0140, 0.0002]	–0.0039 [–0.0118, 0.0039]
Random – PC	+ 0.0413 [0.0292, 0.0533]	+ 0.0231 [0.0140, 0.0327]	– 0.0073 [–0.0137, –0.0013]	–0.0037 [–0.0115, 0.0041]
Random – STDP	+ 0.0267 [0.0154, 0.0381]	+ 0.0142 [0.0051, 0.0238]	– 0.0073 [–0.0138, –0.0010]	–0.0048 [–0.0123, 0.0028]
BP – FA	+ 0.0138 [0.0092, 0.0186] ^P	+ 0.0087 [0.0053, 0.0125]	+0.0044 [–0.0011, 0.0102]	+0.0016 [–0.0041, 0.0075]
BP – PC	+ 0.0104 [0.0065, 0.0149]	+ 0.0066 [0.0035, 0.0101]	+0.0037 [–0.0017, 0.0097]	+0.0017 [–0.0036, 0.0072]
BP – STDP	– 0.0042 [–0.0079, –0.0004]	–0.0023 [–0.0050, 0.0007]	+0.0037 [–0.0017, 0.0097]	+0.0006 [–0.0043, 0.0056]
FA – PC	– 0.0033 [–0.0054, –0.0011] ^P	– 0.0021 [–0.0037, –0.0005]	–0.0007 [–0.0032, 0.0018]	+0.0001 [–0.0041, 0.0044]
FA – STDP	– 0.0179 [–0.0229, –0.0130] ^P	– 0.0110 [–0.0146, –0.0077]	–0.0007 [–0.0042, 0.0030]	–0.0009 [–0.0050, 0.0029]
PC – STDP	– 0.0146 [–0.0186, –0.0105]	– 0.0089 [–0.0119, –0.0062]	+0.0000 [–0.0030, 0.0030]	–0.0011 [–0.0045, 0.0022]

Table 11: Accuracy printed in v2 (training batches, original runs) and CIFAR-10 test accuracy of the repaired models (seed 0).

Condition	v2: training accuracy (%)	v4: test accuracy (%)
Backprop	82.4	75.0
STDP	63.2	57.4
Predictive Coding	56.6	56.7
Feedback Alignment	39.0	38.8
Random Weights	10.0	10.0

Table 12: [Errata of v2: printed value and the value of its source \(original analysis\)](#)

Location in v2	Quantity	printed	correct
Abstract, §4.2, Tab. 2, §5	Random, V1	0.076	0.075
Abstract, §4.2, Tab. 2, §5	BP, V1	0.034	0.033
Tab. 2	BP, V2	0.019	0.018
Tab. 2	BP, V2, CI upper	.023	.022
Abstract, §1, §4.2, §5	$\Delta\rho$ Random – BP, V1	+0.044	+0.042
§4.8	Gabor peakedness, BP	2.00	1.99
§4.5	seed SD range, V1/V2	0.003–0.007	0.003–0.011
§4.4	FDR-significant comparisons of 40	30	26
§4.4	largest FDR p at V1/V2	$p \leq 0.014$	0.0144